Research Summary

Priscilla Kyei Danso
Department of Computer Science, Stony Brook University, New York, USA
priscillakyeidanso@gmail.com

Research Summary: Verifier-Controlled Regulatory Compliance

My research explores how large language models (LLMs) can assist with regulatory compliance while keeping policy decisions grounded in machine-checkable rules. This work grew out of a research direction originally proposed by my advisor, Omar Chowdhury, in a proposal for *COMPLIANCEGPT: An LLM-Assisted Regulatory Compliance Verifier*. The proposal envisioned combining LLMs with formal methods to make complex privacy regulations accessible through natural-language interaction while retaining traceability and verifiability. Although the proposed project was not funded, its central research problem became the foundation for my subsequent work. As I began developing and experimentally studying the proposed LLM–formal-verification pipeline, a more fundamental problem emerged: **the formal verifier can only reason about the evidence supplied to it.** This led to my current work, *ATHENA*.

From the COMPLIANCEGPT Research Direction to ATHENA

The original COMPLIANCEGPT proposal considered an architecture in which an LLM would translate a natural-language compliance question into a structured representation, a symbolic compliance checker would evaluate that representation against a formalized regulation, and another language model would explain the resulting decision. I used this proposed architecture as the starting point for exploring LLM-assisted HIPAA compliance.

My experiments investigated multiple ways of combining language models with a formal HIPAA policy, including direct LLM judgment, retrieval-augmented generation, multi-pass extraction, and agentic approaches. These experiments exposed a limitation that was not adequately addressed by the original architecture: **the verifier does not control the construction of its own evidence.** If an LLM extracts a complete set of facts before verification, an omitted fact becomes invisible to the downstream policy engine. The symbolic verifier can check the supplied facts, but it cannot determine whether an omitted fact was actually necessary for the decision.

This observation shifted the research question. Rather than asking only how an LLM can translate natural-language compliance questions into a formal policy representation, I began asking *who should control which evidence is acquired for a policy decision?* This became the foundation of ATHENA.

ATHENA: Verifier-Controlled Fact Finding

ATHENA answers regulatory permissibility questions by giving the formal policy verifier control over evidence acquisition. Rather than asking an LLM to extract every potentially relevant fact up front, ATHENA begins with structural facts grounded from the scenario and evaluates a machine-checked representation of the HIPAA Privacy Rule.

When the current facts do not determine a policy outcome, the verifier examines its remaining proof obligations and identifies a predicate whose value can still affect the decision. An evidence oracle—implemented using an LLM in our experiments—is then asked a narrowly scoped question about that predicate. The oracle returns `TRUE`, `FALSE`, or `UNKNOWN`. The verifier incorporates the response into the proof state and reevaluates the policy. This continues until the policy establishes a permitted or denied outcome, or until a material predicate cannot be established.

The key architectural distinction is that the LLM supplies evidence but does not decide which regulatory conditions matter. The verifier determines what evidence is relevant to the current proof. Importantly, `UNKNOWN` remains unresolved rather than being interpreted as `FALSE`. Thus, missing evidence is not silently converted into negative evidence merely because an evidence source failed to establish a fact.

Evaluation and Findings

I evaluated ATHENA on real-world HIPAA disclosure scenarios and compared it with direct LLM judgment and batch extract-then-verify architectures. The primary benchmark contains 132 scenarios, with a fixed scored subset used for architecture comparisons. Seven language models are evaluated, ranging from small open-weight models to large proprietary models. A separate 30-scenario gold benchmark with independently verified predicate-level ground truth is used to validate the verifier's proof-search behavior independently of language-model factual extraction.

The central finding is that **verifier-controlled evidence acquisition changes the failure mode of LLM-assisted compliance**. Across the evaluated models, ATHENA commits zero false permits on the scored benchmark. This does not mean that the evidence oracle is always correct: an incorrect oracle response can still affect a decision. Rather, the verifier constrains which policy conditions the oracle can provide evidence about and preserves uncertainty when material evidence cannot be established.

This reliability comes with an explicit coverage tradeoff. Batch extraction systems tend to produce decisions for more scenarios, but an omitted predicate can silently alter the verifier's input. ATHENA instead returns UNRESOLVED when a material predicate cannot be established. Its lower coverage therefore reflects an explicit refusal to manufacture missing evidence rather than a conventional accuracy improvement.

The experiments also show that verifier-controlled fact finding can be relatively efficient. Because the verifier requests evidence only while a predicate remains material to the current proof, many decisions require only a small number of predicate-level queries. The same evidence interface can accommodate different sources, including an LLM, a human reviewer, or another automated evidence system.

Future Research

My next research direction is to investigate whether verifier-controlled compliance reasoning can extend beyond a single regulatory framework. ATHENA currently provides a controlled setting in which to study evidence acquisition under one formalized regulation. Real-world data practices, however, may fall under multiple regulatory regimes simultaneously.

One scenario I am exploring is a person from the European Union receiving medical care at a hospital in the United States. Depending on the entities involved, the individuals concerned, and the processing activities, requirements from the GDPR may become relevant alongside HIPAA. These frameworks have different scopes, definitions, and conditions, raising questions about how a compliance system should represent regulatory applicability and reason over overlapping requirements.

This direction is still exploratory; I have not yet built a multi-regulatory verifier. The immediate research questions are how to represent applicability across regulations, how evidence should be shared between policy-specific verifiers, and how a system should reason when multiple regulatory frameworks impose different conditions on the same data operation.

More broadly, my research goal is to develop AI systems in which language models provide useful natural-language reasoning and evidence while **machine-checkable policies retain authority over consequential decisions**. ATHENA is the first concrete realization of this direction, using privacy regulation as a setting in which the boundary between language-model inference, evidence acquisition, and formal decision making can be studied directly.